

CREStE-Nano — end-to-end architecture

monocular RGB + GPS → BEV traversability → MPPI → PWM · ~\$500 BOM · all 13 ROS 2 nodes on a single Jetson Orin Nano

Perception · 5 FPS on Jetson Orin Nano

Learned reward model

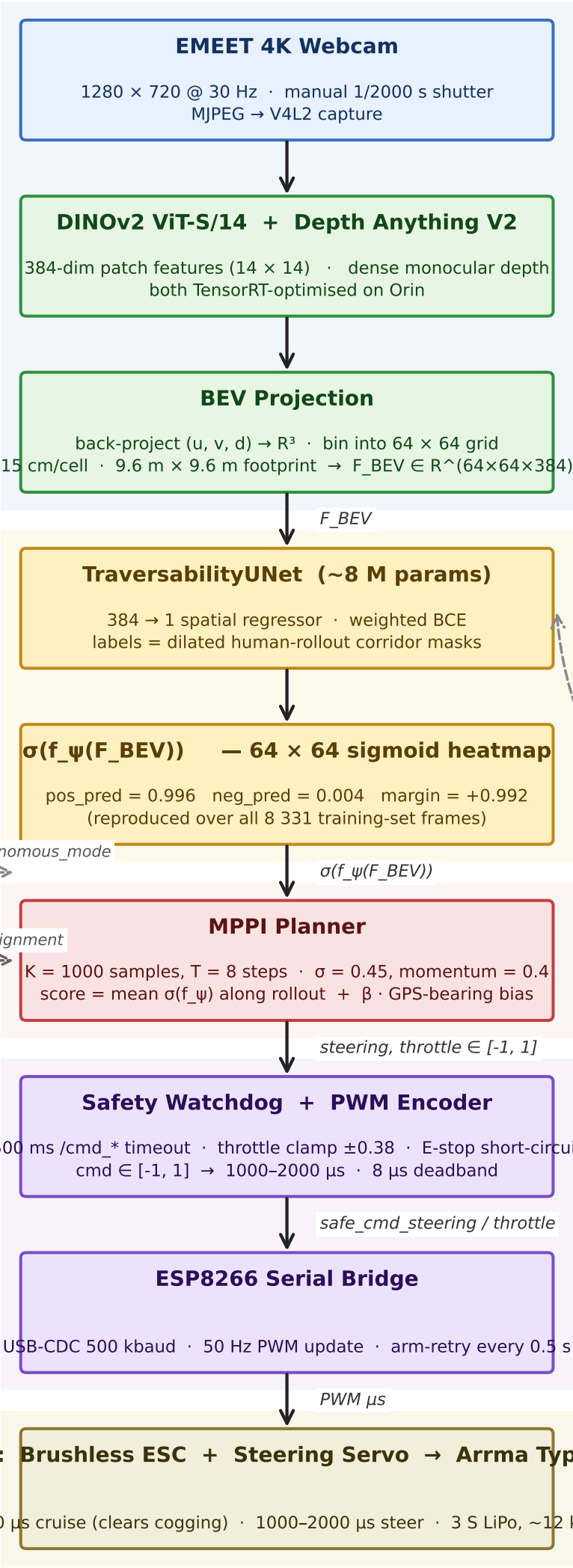
Web Dashboard
phone-friendly UI
E-stop / waypoints / log

GPS (u-blox @ 10 Hz)
waypoint bearing
biases MPPI score

Planning

Safety + low-level control

Hardware



/autonomous_mode

+β·alignment

F_BEV

σ(f_ψ(F_BEV))

steering, throttle ∈ [-1, 1]

safe_cmd_steering / throttle

PWM μs

future feedback